Statistical Minimax Rates in Nonparametric Hypothesis Testing vs. Estimation

In this note, we explain why nonparametric estimation is a harder problem than hypothesis testing, in the sense of minimax rate of statistical convergence. We will follow some definitions and statements from Giné and Nickl (2015). Intuitively speaking, in the spirit of Kolmogorov-Smirnov (KS) nonparametric testing[1], we can use an estimator to construct a testing procedure that achieves at least the same rate as estimation. In particular, any estimator $\hat{f}_n$ with risk $\mathbb{E}d(\hat{f}_n, f) \leq r_n$ (where r_n is the rate) immediately yields a plug-in test statistic $T_n = d(\hat{f}_n, f_0)$ that distinguishes the null hypothesis f_0 from $\{f : d(f, f_0) \geq cr_n\}$ where c is a separation constant that we will later make clear. Hence, testing can never be harder than estimation regardless of the underlying geometry. Furthermore, it can be shown that under the L^2 norm over Sobolev/Besov balls, estimation is strictly harder in terms of minimax rate than testing.

1 Definitions

The following defines the minimax rate ρ_n for distinguishing between a null hypothesis and a separated alternative under distance d.

Definition 1.1 (Definition 6.2.1 in Giné and Nickl (2015). Minimax d-Separation Rate for Hypothesis Testing.)**.**

Let $(\mathbb{P}_f : f \in \mathcal{F})$ be probability laws where f indexes the model, d be a metric on $\mathcal{F}$, and $H_0 \subset \mathcal{F}$ be a null hypothesis. Let $\Psi_n(Y^{(n)}) \to \{0, 1\}$ be the statistical test, which maps data observations $Y^{(n)}$ to reject or accept. For the sequence $(r_n)_n$ of non-negative real numbers, define the separated alternative:

$$H_1(d, r_n) = \left\{ f \in \mathcal{F} : \inf_{g \in H_0} d(f, g) \geq r_n \right\}.$$

A sequence $(\rho_n)_n$ of non-negative numbers is the minimax d-separation rate for testing H_0 versus H_1 if:

[1] The KS statistic quantifies distance between an two CDFs, by the test statistic $T_n = \|F_n - F_0\|_\infty$.

1. *(Feasibility). For every $\alpha > 0$, there exists a test Ψ_n such that for all n sufficiently large,*

$$\sup_{f \in H_0} \mathbb{E}_f \Psi_n + \sup_{f \in H_1(d,\rho_n)} \mathbb{E}_f(1 - \Psi_n) \leq \alpha.$$

2. *(Asymptotic optimality). For any sequence $r_n = o(\rho_n)$,*

$$\liminf_{n \to \infty} \inf_{\Psi_n} \left[\sup_{f \in H_0} \mathbb{E}_f \Psi_n + \sup_{f \in H_1(d,r_n)} \mathbb{E}_f(1 - \Psi_n) \right] > 0.$$

The first statement says that reliable separation is statistically possible. While this allows for any overly conservative sequence where distance is arbitrarily large, the second statement ensures that ρ_n is the tightest possible separation boundary, requiring that any strictly faster rate $r_n = o(\rho_n)$ makes the testing problem impossible since the worst case type 1 and type 2 errors cannot be driven to zero by any test procedure.

The next definition is for the minimax rate for estimation, where the statistical experiment is indexed again by the sample size.

Definition 1.2 (Definition 6.3.1 in Giné and Nickl (2015). Minimax Rate of Estimation in d-Risk.)**.**

Let $\mathcal{F}$ be a subset of a metric space (S, d). A sequence r_n is the minimax rate of estimation in d-risk over $\mathcal{F}$ if:

1. *(Feasibility). There exists an estimator $\hat{f}_n : \mathcal{Y}_n \to S$ and a constant $\overline{C} < \infty$ such that for all $n \in \mathbb{N}$ sufficiently large,*

$$r_n^{-1} \sup_{f \in \mathcal{F}} \mathbb{E}_f d(\hat{f}_n, f) \leq \overline{C}.$$

2. *(Asymptotic Optimality). There exists a constant $\underline{C} > 0$ such that*

$$\liminf_{n \to \infty} r_n^{-1} \inf_{\tilde{f}_n} \sup_{f \in \mathcal{F}} \mathbb{E}_f d(\tilde{f}_n, f) \geq \underline{C}$$

where the infimum extends over all measurable estimators $\tilde{f}_n : \mathcal{Y}_n \to S$.

The first condition states that the worst-case risk of $\hat{f}_n$ decays at least as fast as r_n uniformly over $\mathcal{F}$. The second condition guarantees that r_n is sharp, in the sense

that no estimator can achieve a strictly faster convergence rate uniformly over $\mathcal{F}$.

2 Testing is at most as hard as Estimation

The following proposition shows that testing is at most as hard as estimation. The idea is to use an estimator as a plug-in to the hypothesis test and hence achieve the same rate.

Let $\mathcal{F}$ be a statistical model contained in a metric space (S, d). Suppose there exists an estimator $\hat{f}_n : \mathcal{Y}_n \to S$ satisfying:

$$\sup_{f \in \mathcal{F}} \mathbb{E}_f d(\hat{f}_n, f) \leq r_n$$

for a sequence of non-negative real numbers $(r_n)_{n \in \mathbb{N}}$. To test $H_0 : f = f_0$ for some $f_0 \in \mathcal{F}$, define the test statistic and test for $0 < \alpha < 1$:

$$T_n := d(\hat{f}_n, f_0), \qquad \Psi_n := \mathbb{1}\left\{ T_n > \frac{2r_n}{\alpha} \right\}.$$

Note that the threshold $\frac{2r_n}{\alpha}$ comes directly from equal error splitting ($\frac{\alpha}{2}$ for Type I, $\frac{\alpha}{2}$ for Type II) via Markov's inequality.[2]

Proposition 1 (Proposition 6.2.2 in Giné and Nickl (2015))**.** *Let $f_0 \in \mathcal{F}$ and $\alpha > 0$ be given, and consider testing the hypotheses*

$$H_0 : f = f_0 \quad \textit{vs.} \quad H_1 \subset \mathcal{F} \cap \{f \in \mathcal{F} : d(f, f_0) \geq c r_n\}, \quad c \geq \frac{4}{\alpha}$$

Then, for Ψ_n defined above, for every $n \in \mathbb{N}$:

$$\mathbb{E}_{f_0} \Psi_n + \sup_{f \in H_1} \mathbb{E}_f (1 - \Psi_n) \leq \alpha$$

Proof. The proof is given in the book. It follows from Markov's inequality and the triangle inequality. ■

Here, recall that $H_0 = \{f_0\}$ and $H_1(d, cr_n) = \{f \in \mathcal{F} : d(f, f_0) \geq c r_n\}$. From this

[2]For example for the Type I error, $\mathbb{P}_{f_0}(d(\hat{f}_n, f_0) > \tau) \leq \frac{\mathbb{E}_{f_0} d(\hat{f}_n, f_0)}{\tau} \leq \frac{r_n}{\tau} \overset{\text{set}}{=} \frac{\alpha}{2} \implies \tau = \frac{2r_n}{\alpha}$.

proposition, setting the separation boundary to $c(\alpha)r_n$ (with $c(\alpha) = 4/\alpha$) yields

$$\sup_{f \in H_0} \mathbb{E}_f \Psi_n + \sup_{f \in H_1(d, c(\alpha)r_n)} \mathbb{E}_f(1 - \Psi_n) \leq \alpha$$

matching the feasibility condition in definition 1.1. Therefore we know that

$$\rho_n \lesssim r_n.$$

Hence, estimation is at least as hard as testing in the sense of minimax rates ($\rho_n \lesssim r_n$), regardless of the underlying metric d or parameter space $\mathcal{F}$.

3 Estimation is strictly harder in L^2

We now show that under the L^2-metric over Besov balls (see Appendix for definition), we can achieve a faster rate of convergence under testing than estimation. There are many possible applications, but one is for densities, as follows.

Consider a metric space $(S, d) = (L^2([0,1]), \|\cdot\|_2)$. For smoothness $r > 0$, radius $B > 1$, and $1 \leq q \leq \infty$, define the parameter space of bounded probability densities:

$$\mathcal{F} = \left\{ f : [0,1] \to [0,\infty) \ : \ \int_0^1 f(x)\,dx = 1, \ \|f\|_{B^r_{2q}([0,1])} \leq B, \ \|f\|_\infty \leq B \right\}$$

The minimax separation rate for testing uniformity over $\mathcal{F}$ (i.e. all observations are drawn from a uniform distribution) is given by the following theorem.

Theorem 1 (Theorem 6.2.9 in Giné and Nickl (2015))**.** *For any $r > 0$, $B > 1$, and $1 \leq q \leq \infty$, the minimax separation rate for testing*

$$H_0 : f = 1 \quad \textit{vs.} \quad H_1(r_n) : f \in \mathcal{F} \cap \{\|f - 1\|_2 \geq r_n\}$$

in the sense of Definition 1.1 based on i.i.d. observations $X_1, \ldots, X_n \sim f$ is equal to

$$\rho_n = C_1 n^{-\frac{r}{2r+1/2}}$$

where $C_1 > 0$ depends only on r, B, and the test level α.

Proof. The proof is given in the book. It relies on Wavelet basis construction and

certain concentration inequalities. ■

In contrast, the minimax rate for estimating f under the identical model and metric is characterized by the following.

Theorem 2 (Theorem 6.3.9 in Giné and Nickl (2015), $p = 2$)**.** *For any $r > 0$, $B > 1$, and $1 \leq q \leq \infty$, the minimax rate of estimation over $\mathcal{F}$ in L^2-risk in the sense of Definition 1.2 based on i.i.d. observations $X_1, \ldots, X_n \sim f$ is equal to*

$$r_n = C_2 n^{-\frac{r}{2r+1}}$$

where $C_2 > 0$ depends only on r and B.

Proof. The proof is given in the book. ■

It can be seen by comparing the rates in the above two theorems that estimation has a slower minimax rate of convergence than hypothesis testing.

4 Appendix: Besov Spaces

The Besov space is a generalization of Sobolev spaces, and is a type of Banach space. I provide a definition as follows, following Chapter 4 in Giné and Nickl (2015).

Let $A \subseteq \mathbb{R}$ be an interval (typically $A = [0, 1]$ or $\mathbb{R}$). For smoothness index $s > 0$ and integrability indices $1 \leq p, q \leq \infty$, the **Besov space** $B^s_{pq}(A)$ is defined as:

$$B^s_{pq}(A) := \begin{cases} \left\{ f \in L^p(A) : \|f\|_{B^s_{pq}(A)} < \infty \right\}, & 1 \leq p < \infty \\ \left\{ f \in C_u(A) : \|f\|_{B^s_{\infty q}(A)} < \infty \right\}, & p = \infty \end{cases}$$

where $C_u(A)$ denotes the space of bounded, uniformly continuous functions on A.

To define the **Besov norm**, there are many equivalent ways, but one route is by **moduli of smoothness**. Fix an integer $r > s$. For $h > 0$, the r-th order difference operator $\Delta^r_h = (\tau_h - \mathrm{id})^r$ (with translation $\tau_h f(x) = f(x + h)$) is given by:

$$\Delta^r_h(f)(x) = \sum_{k=0}^{r} \binom{r}{k} (-1)^{r-k} f(x + kh)$$

with the convention that $\Delta_h^r(f)(x) \equiv 0$ whenever $x + rh \notin A$. For $f \in L^p(A)$, the r-th **modulus of smoothness** in $L^p(A)$ is

$$\omega_r(f, t, p) = \sup_{0<h\leq t} \|\Delta_h^r(f)\|_{L^p(A)}, \quad t > 0.$$

Then, the **Besov norm** is defined as:

$$\|f\|_{B^s_{pq}(A)} := \|f\|_{L^p(A)} + |f|_{B^s_{pq}(A)}$$

where the **Besov seminorm** $|f|_{B^s_{pq}(A)}$ is:

$$|f|_{B^s_{pq}(A)} := \begin{cases} \left(\int_0^\infty \left[\dfrac{\omega_r(f, t, p)}{t^s} \right]^q \dfrac{dt}{t} \right)^{1/q}, & 1 \leq q < \infty \\ \sup\limits_{t>0} \dfrac{\omega_r(f, t, p)}{t^s}, & q = \infty \end{cases}.$$

Finally, for any radius $B > 0$, the **Besov ball** of radius B, denoted $\mathcal{B}^s_{pq}(B)$, is the closed norm ball centered at zero in $B^s_{pq}(A)$,

$$\mathcal{B}^s_{pq}(B) := \left\{ f \in B^s_{pq}(A) : \|f\|_{B^s_{pq}(A)} \leq B \right\}.$$

References

Giné, E. and R. Nickl (2015): *Mathematical Foundations of Infinite-Dimensional Statistical Models*, vol. 40 of *Cambridge Series in Statistical and Probabilistic Mathematics*, Cambridge University Press.